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MILITARY SPECIFICATION

GYROSCOPE ASSEMBLY, RATE SBU-13/A

1. SCOPE

1.1 This specification covers the requirements for one type of rate gyroscope assembly, designated SBU-13/A.

*2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-4150	Case, Carrying, Water-Vaporproof
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods

FSC 6610

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MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS27473	Connector, Plug, Electric, Crimp Type, Straight, Bayonet Coupling Series Li-a
MS27478	Connector, Receptacle, Electric, Solder Type, Solder Mounting Hermetic Seal, Pins, Bayonet Coupling Series Li-a

(Copies of documents required by suppliers in connection with specific procurement functions may be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the gyroscope assembly except where magnetic materials are essential.

3.3.2 Protective treatment. When materials are used in the construction of the gyroscope assembly that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided. Components contained within hermetically sealed containers shall be considered to be adequately protected against deterioration.

3.3.3 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.4 Corrosive fumes. The materials, as installed in the gyroscope assembly and under the service conditions specified herein, shall not liberate deleterious fumes.

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3.3.5 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met during storage or normal service.

* 3.3.5.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4 Design and construction. The gyroscope assembly shall be designed to sense rates about the roll, pitch, and yaw axes and shall be sufficiently rugged to meet the requirements specified herein. It shall be so designed and constructed that no parts will work loose in service and shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 The gyroscope assembly shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases without the use of special tools unless they are approved by the procuring activity.

3.4.2 Rubbing parts. The gyroscope assembly shall not incorporate rubbing parts of components which would be subject to wear (except for bearings.)

* 3.4.3 Mechanism. The gyroscope assembly shall sense and measure the three principle angular motions (pitch, roll, and yaw) of the aircraft and shall provide this information to the remotely located converter (ECU-66/A, -67/A, -68/A, -69/A). The gyroscope assembly shall be located normally at the aircraft center of gravity.

3.4.4 Shock mounts. Shock mounts shall not be used between the gyroscope assembly and the airframe.

* 3.4.5 Maintainability program. The contractor shall establish a maintainability program which shall include but not be limited to: Defining levels (line, shop, and depot) of support required to maintain the gyroscope assembly, predicting expected failure modes, and defining equipment required to repair any failed components.

3.4.5.1 Maintainability. The following maintainability requirements shall be considered in the design of the gyroscope assembly:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

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c. Rapid and positive indentification of the replaceable defective part, assembly or component

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

3.4.6 Reliability. The gyroscope assembly shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,500 hours based on a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.4.6.1 Longevity. The gyroscope assembly shall have an operating life span of not less than 3,000 hours before the equipment consistently fails to meet the specified MTBF index.

3.4.7 Damping. The gyroscope assembly damping shall be of a symmetrical non-gaseous fluid-damped design and shall be temperature compensated without the use of heaters.

3.5 Performance. The gyroscope assembly shall be capable of satisfactory operation when subjected to the following conditions or any combination thereof:

a. Temperature - ambient operating temperatures ranging from -54° to +71°C and operation for 30 minutes at 95°C

b. Humidity - relative humidity up to 95 percent

c. Vibration

(1) Error - vibration at a double amplitude of 0.060 inch through a frequency range of 10 to 55 Hz

(2) Failure - acceleration of 10g through a frequency range of 74 to 500 Hz and 1g from 5 to 23 Hz

d. Acceleration - acceleration forces up to 20g

e. Altitude - pressures ranging from 30 inches Hg (sea level) down to 0.82 inch Hg (approximately 80,000 feet)

f. Fungus - exposure to fungus enzymes

g. Salt fog - exposure to salt fog for a minimum period of 168 hours

h. Dust - exposure to blowing fine sand and dust particles

i. Temperature shock - temperature shock caused by an abrupt temperature drop from +85° to -40°C

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j. Shock - dynamic shock at accelerations up to 20g for a nominal duration of 11 milliseconds

k. Acoustical noise - exposure to acoustical sound fields characteristic of aircraft.

* 3.5.1 Electromagnetic interference and susceptibility. The gyroscope assembly shall meet the electromagnetic interference and susceptibility requirements of MIL-STD-461 for class A1 equipment.

3.5.2 Thermal shock. The case shall show no evidence of leakage when subjected to alternate immersions in tap water at a temperature of 85°C and then in tap water at a temperature of 5°C.

3.5.3 Leak rate. The gyroscope shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 1 atmosphere.

3.5.4 Ranges and accuracies. Ranges and accuracies of the gyroscope assembly shall be in accordance with table I.

*

TABLE I. Ranges and Accuracies

	Roll	Pitch	Yaw
Full Range	±250°/sec	±50°/sec	±50°/sec
Linearity	0.75°/sec at 0°/sec 1.05°/sec at 45°/sec 2.50°/sec at 90°/sec 3.75°/sec at 250°/sec	0.02°/sec at 0°/sec 0.31°/sec at 15°/sec 0.48°sec at 30°/sec 0.75°/sec at 50°/sec	0.08°/sec at 0°sec 0.31°/sec at 15°/sec 0.75°/sec at 30°/sec 0.75°/sec at 50°/sec
Threshold	0.05°/sec	0.02°/sec	0.015°/sec
Hysteresis	0.05 percent of input rate	0.05 percent of input rate	0.05 percent of input rate
Repeatability	0.2 percent of full scale	0.2 percent of full scale	0.2 percent of full scale

3.5.5 Output signal. Each axis (roll, pitch, and yaw) output signal shall be ±5V dc at the maximum input rate.

* 3.5.6 Frequency response. The frequency response of the transmitter in any axis shall be flat within ±1.5 percent of static response from 0 through 6 Hz at room temperature and ±3.0 percent over the specified temperature range, and shall be down 18 db or more at 12 Hz. Thereafter, the rolloff shall be 30 db/octave.

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*3.5.7 Insulation resistance. The gyroscope assembly shall withstand 200V dc applied between each external connection of mutually insulated circuits and between each external connection of mutually insulated circuits and the case. The insulation resistance shall be a minimum of 5 megohms. This requirement does not apply to any circuit whose impedance to the case is designed to be less than 5 megohms.

*3.5.8 Electrical transients. The gyroscope shall be designed to withstand repeated applications of transient spikes and surge voltages applied between pins 2 and 10 (power input). The spike voltages shall be ±600V, ±200V, +80V, and +73V.

3.5.9 Warmup. The gyroscope shall not require a warmup period of more than 90 seconds under any temperature conditions specified herein.

3.5.10 Overrange. The gyroscope assembly shall withstand at least 50 percent overrange in any sensing parameter without change in calibration.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.7 Electric circuit

3.7.1 Power source. The gyroscope assembly shall be designed to operate from a 28V d-c power source as defined in MIL-STD-704.

3.7.2 Power inversion. The 28V d-c power shall be converted to a regulated a-c voltage by means of a static inverter within the instrument case.

*3.7.3 Power consumption. Power consumption of the gyroscope assembly shall not exceed 25 watts starting and 15 watts running.

*3.7.4 Undervoltage. The gyroscope assembly shall not be damaged when operated by power reduced to 21V dc, and shall automatically resume normal operation when the voltage returns to its nominal value.

3.8 Spin motor. The spin motor shall not incorporate slip rings, contacts, brushes, or any similar element that may fail due to wear incurred in normal operation.

3.9 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists shall not be used unless approved by the procuring activity.

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3.10 Case. The case shall be made of sturdy material which shall be uniform in texture and shall be covered in durable dull black. It shall be gasket sealed and of such design that the internal mechanism may be removed from the case as a unit, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity. The seal shall be accomplished in such a manner that it will not be dependent upon materials that might be affected by the action of any atmosphere to which the gyroscope assembly may be subjected. The case shall be so rugged that it will not become distorted to such extent as to degrade performance of the gyroscope assembly. It shall be marked to indicate the location of the precession axis and the forward direction for proper installation in aircraft.

3.10.1 Filling medium. The filling medium in the gyro subassembly case shall be at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.11 Time totalizing meter. A 10,000-hour time totalizing meter conforming to MS17322 shall be provided and so mounted within the case that the counter display will be readable on the connector end of the gyroscope assembly. The meter shall be capable of operating in temperatures ranging from -55° to +85°C, may be either round or square, and may be with or without a flange. The meter shall not be removable from outside the gyroscope assembly case.

3.12 Weight. The total weight of the completely assembled gyroscope assembly, less mounting and cables, shall not exceed 4 pounds maximum.

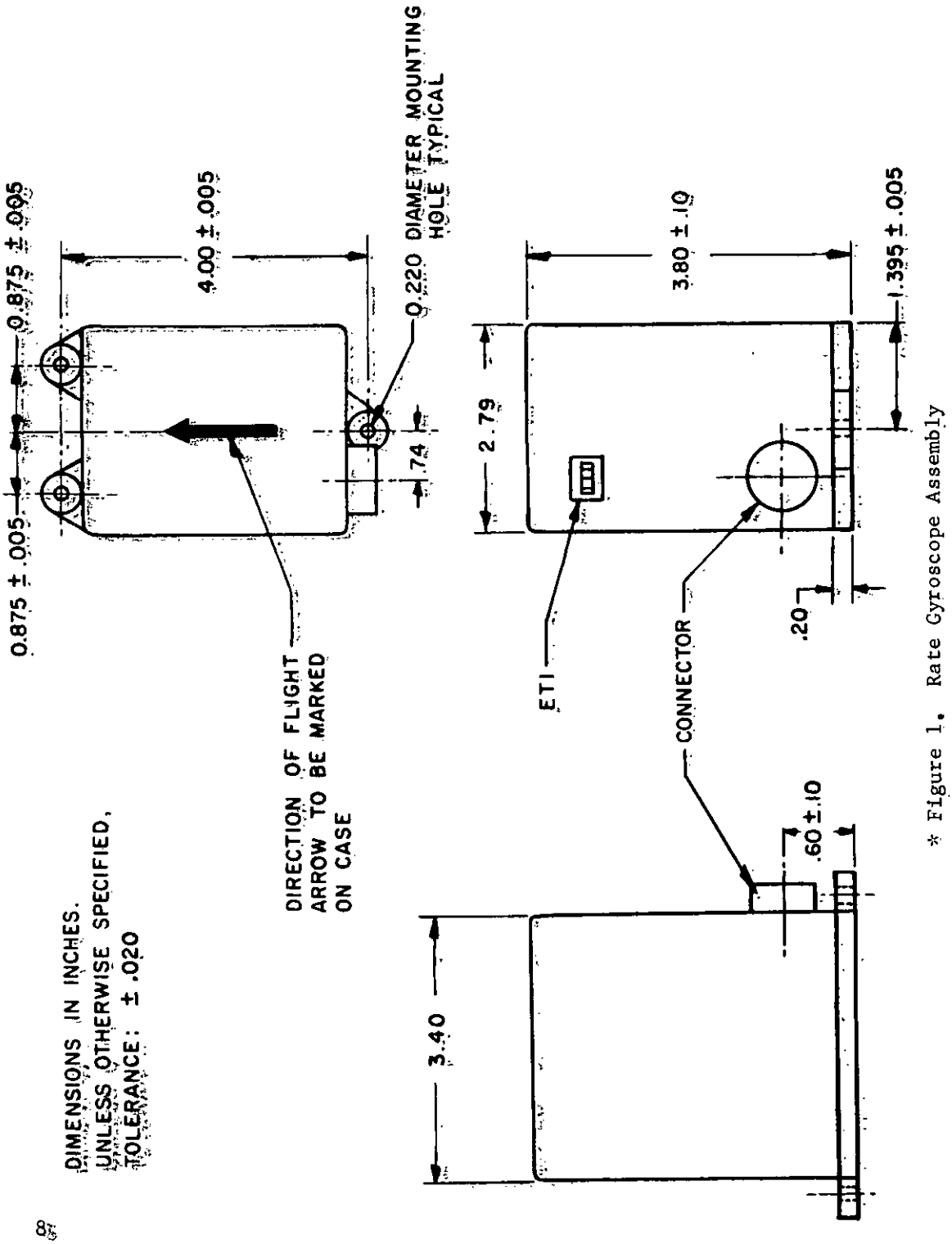
*3.13 Size. The overall dimensions of the gyroscope assembly shall be in accordance with figure 1.

*3.14 Electrical connector. An MS27478Y10D13P electrical connector shall be provided as part of the gyroscope assembly and shall mate with an MS27473E10A13S connector. The pin designation shall be in accordance with table II.

3.15 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454, requirement 5.

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* Figure 1. Rate Gyroscope Assembly

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* Table II. Connector Pins and Functions

Pin	Function
1	Case ground
2	+28V
10	-28V
5	Roll signal
6	Pitch signal
7	Yaw signal
4	Roll ground
12	Pitch ground
13	Yaw ground
3	Spare
8	Spare
9	Spare
11	Spare

3.17 Finishes and protective coatings

3.17.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.17.1.1 Small holes and case inserts need not be anodized.

3.17.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.17.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 shall be used in lieu of anodizing.

3.17.1.4 Casting containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.17.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.17.1.6 Parts enclosed in a hermetically sealed container need not be anodized or plated.

3.17.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

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3.17.1.8 When necessary for electrical bonding, parts need not be anodized.

3.17.2 Steel parts. Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship. The gyroscope assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.19.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the gyroscope assembly, they shall be held or limited accordingly.

3.19.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened without damage or injury to the screw or bolt or threads.

3.19.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.4 Cleaning. The gyroscope assembly shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign materials removed after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of gyroscope assemblies shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the gyroscope assembly shall be tested in its normal operating position.

4.3.3 Voltage input. Unless otherwise specified, the gyroscope assembly shall be tested with 28V dc nominal power.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of 12 gyroscopes representative of the production equipment. Three of the gyroscopes shall be subjected to all the tests except reliability and the other nine gyroscopes shall be subjected to the reliability test only (see 4.6.27). Accept-reject criteria shall be in accordance with figure 2.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

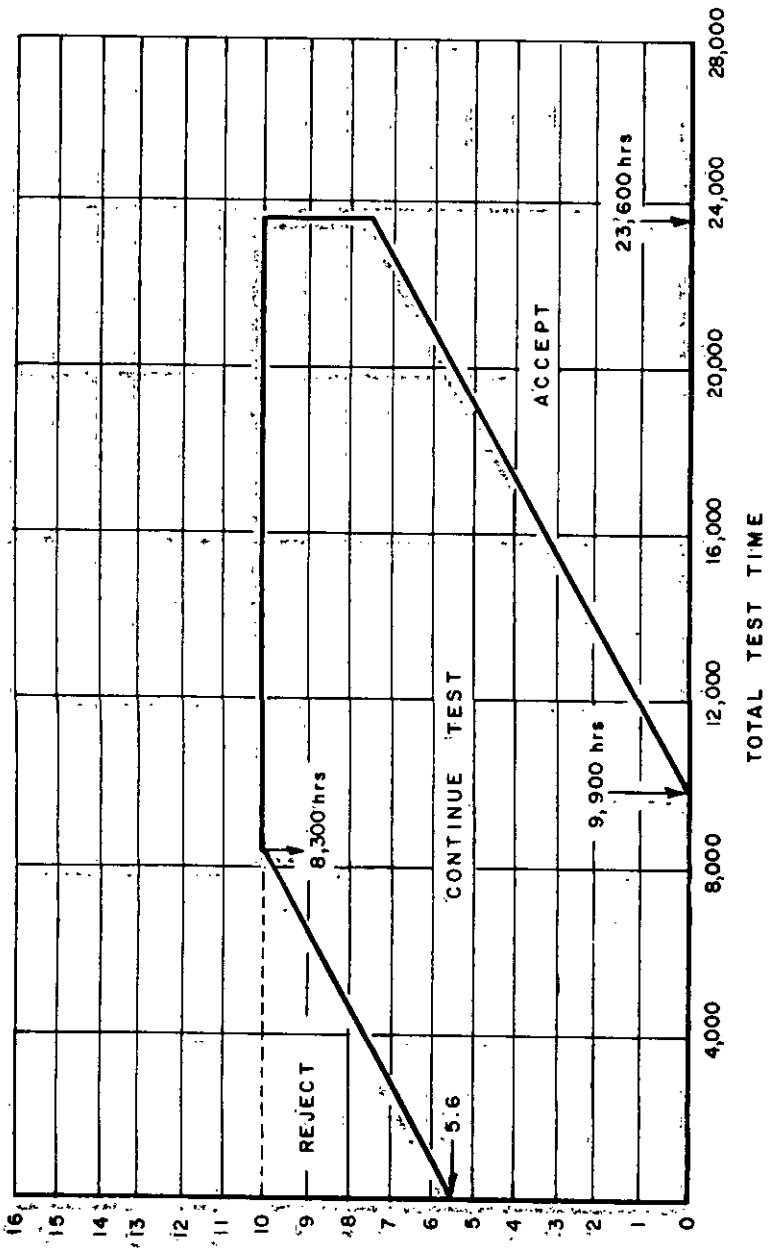
4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

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* Figure 2. Accept, Continue-Test, Reject Criteria

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4.5.1 Individual tests. Each gyroscope assembly shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Burn-in

c. Power consumption

d. Sensitivity

e. Repeatability

f. Resolution and threshold

g. Friction and hysteresis

h. Static balance

i. Overrange

j. Frequency response

k. Insulation resistance.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One gyroscope assembly selected at random from each 100 or fraction thereof on the contract or order shall be subjected to the following tests as described under 4.6:

a. Individual tests (except burn-in)

b. Sensitivity at low temperature

c. Sensitivity at high temperature

d. Voltage extremes

e. Electrical transients

f. Vibration error

g. Sealing.

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4.5.2.2 Sampling plan B. Unless otherwise specified, 3 gyroscopes selected at random from the first 15 on the contract or order shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Electromagnetic interference and susceptibility

c. Temperature shock

d. Temperature-altitude cycling

e. Thermal shock

f. Humidity

g. Fungus

h. Salt fog

i. Dust

j. Acceleration

k. Shock

l. Vibration failure

m. Acoustical noise.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure; but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.5.2.4 Sampling plan C. Unless otherwise specified, five gyroscopes shall be selected at random from each 500 or less produced on the contract or order and tested as specified in 4.6.27 (does not include 4.6.27.1) for a total of 5,000 hours.

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4.5.2.4.1 Acceptance and rejection. Figure 2 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.27. In the event a continue-test decision applies after 5,000 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for a composite accept-reject decision. If the continue-test condition persists, the data from successive production quantities shall be accumulated until 10 failures or 23,600 hours of testing have occurred at which time an accept-reject decision shall be made. In the event of a reject decision, production units shall be considered unacceptable and deliveries shall be stopped immediately. No more deliveries shall be made until further notice from the procuring activity.

4.5.2.4.2 Recording. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall full advise the procuring activity of defects likely to be found and methods of correcting them.

4.5.3.1 Corrective action and retest. Corrective action and retest shall be required for each failure that can be reasonably attributed to poor manufacturing or design practices or which adversely affects mission completion.

4.6 Test methods

4.6.1 Examination of product. The gyroscope assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Burn-in. After completion of examination of product (4.6.1) the gyroscope assembly shall be subjected to a burn-in test. It shall be operated on a scorsby table as described in 4.6.27 for a period of 24 hours in the normal position and at room temperature.

4.6.3 Power consumption. The gyroscope assembly shall be connected for normal operation without power supplied and with the gyro rotors at rest. Power shall be applied and the starting power, measured within 5 seconds from the time power is applied, shall not exceed 25 watts. Ninety seconds subsequent to application of power, the power consumption shall not exceed 15 watts.

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4.6.4 Sensitivity

*4.6.4.1 Roll. The gyroscope assembly shall be properly connected and so mounted on a rate table that it will be sensitive about its roll axis. The following outputs about the roll axis in both clockwise and counterclockwise directions shall be observed when viewing the connector end (positive output for clockwise rotation and negative output for counterclockwise rotation). The yaw and pitch outputs shall remain at 0V ±63 mv and 0V ±60 mv, respectively, for each roll rate specified herein.

Rate of Roll (Degrees Per Second)	Output
0	0V ±15 mv
15	0.300V ±15 mv
30	0.600V ±18 mv
45	0.900V ±21 mv
90	1.800V ±50 mv
180	3.600V ±53 mv
210	4.200V ±75 mv
250	5.000V ±75 mv.

*4.6.4.2 Pitch. The gyroscope assembly shall be properly connected and so mounted on a rate table that it will be sensitive about the pitch axis. The following outputs about the pitch axis in both clockwise and counterclockwise directions shall be observed (positive output for pitch-down rotation and negative output for pitch-up rotation). The yaw and roll outputs shall remain at 0V ±20 mv and 0V ±25 mv, respectively, during each pitch rate.

Rate of Pitch (Degrees Per Second)	Output
0	0V ±20 mv
5	0.500V ±20 mv
15	1.500V ±31 mv
30	3.000V ±75 mv
50	5.000V ±75 mv.

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*4.6.4.3 Yaw. The gyroscope assembly shall be properly connected and so mounted on a rate table that it will be sensitive about the yaw axis. The following output about the yaw axis in both clockwise and counterclockwise directions shall be observed viewing from above (positive output for clockwise rotation and negative output for counterclockwise rotation). The roll and pitch outputs shall remain at 0V ±25 mv and 0V ±30 mv, respectively, for each rate of turn specified herein.

Rate of Turn (Degrees Per Second)	Output
0	0V ±8 mv
5	0.500V ±18 mv
15	1.500V ±31 mv
30	3.000V ±75 mv
50	5.000V ±75 mv

4.6.5 Repeatability

*4.6.5.1 Roll. The gyroscope assembly shall be so mounted on a rate table that it will be sensitive in the roll axis and shall be properly connected with power applied. The rate table shall be rotated at 0°, 30°, 90°, 180°, and 250° per second, consecutively. At each of these rates, the output shall be recorded. This procedure shall be repeated two more times. The reading at each rate shall not vary from the average of the three readings by more than 0.20 percent of full scale or 10 mv.

*4.6.5.2 Pitch. The gyroscope assembly shall be so mounted on a rate table that it will be sensitive in the pitch axis and shall be properly connected with power applied. The rate table shall be rotated at 0°, 15°, 30°, and 50° per second consecutively. At each of these rates, the output shall be recorded. This procedure shall be repeated two more times. The readings at each rate shall not vary from the average of the three readings by more than 0.20 percent of full scale or 10 mv.

*4.6.5.3 Yaw. The gyroscope assembly shall be so mounted on a rate table that it will be sensitive in the yaw axis and shall be properly connected with power applied. The rate table shall be rotated at 0°, 15°, 30°, and 50° per second. At each of these rates, the output shall be recorded. This procedure shall be repeated two more times. The readings at each rate shall not vary from the average of the three readings by more than 0.20 percent of full scale or 10 mv.

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* 4.6.6 Resolution and threshold. Both resolution and threshold shall be tested at a rate of 0.05° per second in the roll axis, 0.02° per second in the pitch axis, and 0.015° per second in the yaw axis so that the perceptible change in output signal will be at least 0.9 mv for the roll axis, 1.75 mv for the pitch axis, and 1.25 mv for the yaw axis.

* 4.6.7 Friction and hysteresis. The gyroscope assembly shall be tested for friction and hysteresis for a minimum period of 15 seconds in both clockwise and counterclockwise directions. The total variation at zero input for each axis at the maximum rates (250° per second for roll, 50° per second for pitch, and 50° per second for yaw) shall not exceed 2.5 mv.

4.6.8 Static balance. The gyroscope assembly shall be mounted upside down in the test fixture and subjected to the tests specified under 4.6.4.

* 4.6.9 Overrange. The gyroscope assembly shall be so mounted on a rate table that it will be sensitive about its yaw axis and shall be properly connected with power applied. The table shall be rotated at a rate of 75° per second for a period of 3 minutes. The gyroscope assembly shall then be positioned so that it will be sensitive about its pitch axis and the table rotated at a rate of 75° per second for a period of 3 minutes after which it shall be positioned so that it is sensitive about its roll axis and the table rotated at a rate of 375° per second for 3 minutes. At the completion of this test, the gyroscope assembly shall pass the tests specified under 4.6.4.

* 4.6.10 Frequency response. The frequency response test shall be conducted at 25°C and at frequencies from static (0.5 Hz or less) through 24 Hz. Characteristics of the frequency response shall be as specified in 3.5.6.

* 4.6.11 Insulation resistance. The insulation resistance shall be measured by applying 200V dc between each external connection of mutually insulated circuits and between each external connection of mutually insulated circuits and the case. The insulation resistance shall be a minimum of 5 megohms. This test shall not apply to any circuit whose impedance to the case is less than 5 megohms.

* 4.6.12 Sensitivity at low temperature. The gyroscope assembly shall be properly connected and placed in a chamber at a temperature of -54° ±2°C for a period of 4 hours. At the end of the 4 hours and while still at the low temperature, 28V dc shall be applied for 10 minutes. The gyroscope assembly shall meet the following:

Rate of Roll Input (degrees per second)	Output at -54°C
0	0V ±19 mv
15	0.300V ±28 mv
30	0.600V ±33 mv
45	0.900V ±38 mv
90	1.800V ±55 mv
180	3.600V ±90 mv
210	4.200V ±90 mv
250	5.000V ±95 mv

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Rate of Pitch Input (degrees per second)	Pitch Output at -54°C
0	0V ±24 mv
5	0.500V ±29 mv
15	1.500V ±41 mv
30	3.000V ±90 mv
50	5.000V ±98 mv

Rate of Turn Input (degrees per second)	Yaw Output at -54°C
0	0V ±24 mv
5	0.500V ±29 mv
15	1.500V ±41 mv
30	3.000V ±90 mv
50	5.000V ±98 mv.

In addition, the gyroscope assembly shall be subjected to the test specified in 4.6.10, except that the frequency response shall be flat to within ±3 percent of static response and the phase linearity shall be ±4° from the best straight line in the band pass.

* 4.6.13 Sensitivity at high temperature. The gyroscope assembly shall be placed in a chamber at a temperature of 71° ±2°C for a period of 4 hours. At the end of the 4 hours and while still at 71° ±2°C, 28V dc shall be applied to the gyroscope assembly. The gyroscope shall meet the following:

Rate of Roll Input (degrees per second)	Roll Output at +71°C
0	0V ±17 mv
15	0.300V ±18 mv
30	0.600V ±20 mv
45	0.900V ±23 mv
90	1.800V ±58 mv
180	3.600V ±60 mv
210	4.200V ±80 mv
250	5.000V ±90 mv

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Rate of Pitch Input (degrees per second)	Pitch Output at 71°C
0	0V ±24 mv
5	0.500V ±25 mv
15	1.500V ±41 mv
30	3.000V ±90 mv
50	5.000V ±90 mv

Rate of Turn Input (degrees per second)	Yaw Output at 71°C
0	0V ±11 mv
5	0.500V ±25 mv
15	1.500V ±35 mv
30	3.000V ±90 mv
50	5.000V ±90 mv.

In addition, the gyroscope assembly shall be subjected to the test specified in 4.6.10 except that the frequency response shall be flat to within ±3 percent of the static response and the phase linearity shall be ±4° from the best straight line in the band pass.

* 4.6.14 Voltage extremes. Voltages of 21 and 31V dc shall be applied to the gyroscope assembly. While each voltage is applied, the tests specified under 4.6.4 shall be repeated, except that all readings shall be +10 and -17 percent of the normal output values at the respective rates of turn for each axis.

4.6.15 Electrical transients. While operating from normal power, the gyroscope shall be subjected to a minimum of 15 applications of each of the following voltage transients. The interval between applications shall be from 1 to 2 seconds. After subjection to these voltage transients, the gyroscope shall meet the tests specified under 4.6.4 and in 4.6.10.

D-C Voltage Transient	Duration
+600, -600	10 microseconds
+200, -200	1 millisecond
+80	50 milliseconds
+73	200 milliseconds.

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*4.6.16 Vibration error. The vibration error test shall be conducted with power applied. The test shall consist of a frequency survey with vibration applied to the longitudinal, lateral, and vertical axes of the gyroscope assembly. The gyroscope assembly shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through a frequency range of 10 to 55 Hz in each of the three mutually perpendicular axes. Duration of the vibration shall be 3 minutes with vibration applied in a linear motion for one survey of the complete frequency range. During this test, the output signal shall not exceed 25 mv, 35 mv, and 25 mv for roll, pitch, and yaw respectively.

4.6.17 Sealing. The case shall be tested for leaks by means of a mass-spectrometer-type of helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi. This test shall be conducted on the case prior to final assembly of the gyroscope assembly.

4.6.18 Electromagnetic interference and susceptibility. The gyroscope assembly shall be subjected to the electromagnetic interference and susceptibility tests specified in MIL-STD-462. The test methods shall be as specified in MIL-STD-461 for class A1 equipment.

4.6.19 Temperature shock. The temperature shock test shall be conducted in accordance with method 503 of MIL-STD-810. After the gyroscope assembly has returned to room temperature, the connectors and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The gyroscope assembly shall then pass the individual tests except 4.6.2.

4.6.20 Temperature-altitude. The gyroscope assembly shall be subjected to a temperature-altitude test in accordance with MIL-STD-810, method 504, class 2, procedure I, except that the altitude shall be 80,000 feet and in step 5, the stabilized temperature shall be maintained for 4 hours. The gyroscope assembly shall operate without sticking. After returning to room temperature and atmospheric pressure, the gyroscope assembly shall pass the individual tests except 4.6.2. It shall operate properly throughout this test and there shall be no damage that would affect subsequent operation.

4.6.21 Thermal shock. The gyro subassembly case shall be immersed alternately in tap water maintained at 85° ±5°C and 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes with not more than 5 seconds elapsing between bath immersions. No damage to the hermetic seal shall occur as a result of this test. Following the immersions, the case shall be subjected to and shall meet the tests specified under 4.6.17.

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4.6.22 Environmental tests. The following environmental tests shall be conducted in accordance with the specified methods and procedures of MIL-STD-810. At the completion of each test, the gyroscope assembly shall pass the individual tests except 4.6.2, 4.6.8, and 4.6.9.

4.6.22.1 Humidity. The humidity test shall be conducted in accordance with method 507, procedure I.

4.6.22.2 Fungus. The fungus test shall be conducted in accordance with method 508, procedure I.

4.6.22.3 Salt fog. The salt fog test shall be conducted in accordance with method 509, procedure I, for a period of not less than 168 hours.

4.6.22.4 Dust. The dust test shall be conducted in accordance with method 510, procedure I.

4.6.23 Acceleration. The gyroscope assembly, not operating, shall be mounted in its normal operating position and subjected to an acceleration of 20g, first along its vertical axis and then along each of the two axes that are perpendicular to the vertical axis and to each other, for a period of 1 minute in each axis. At the conclusion of this test, the gyroscope assembly shall meet the individual tests except 4.6.2, 4.6.8, and 4.6.9. No damage shall result from this test.

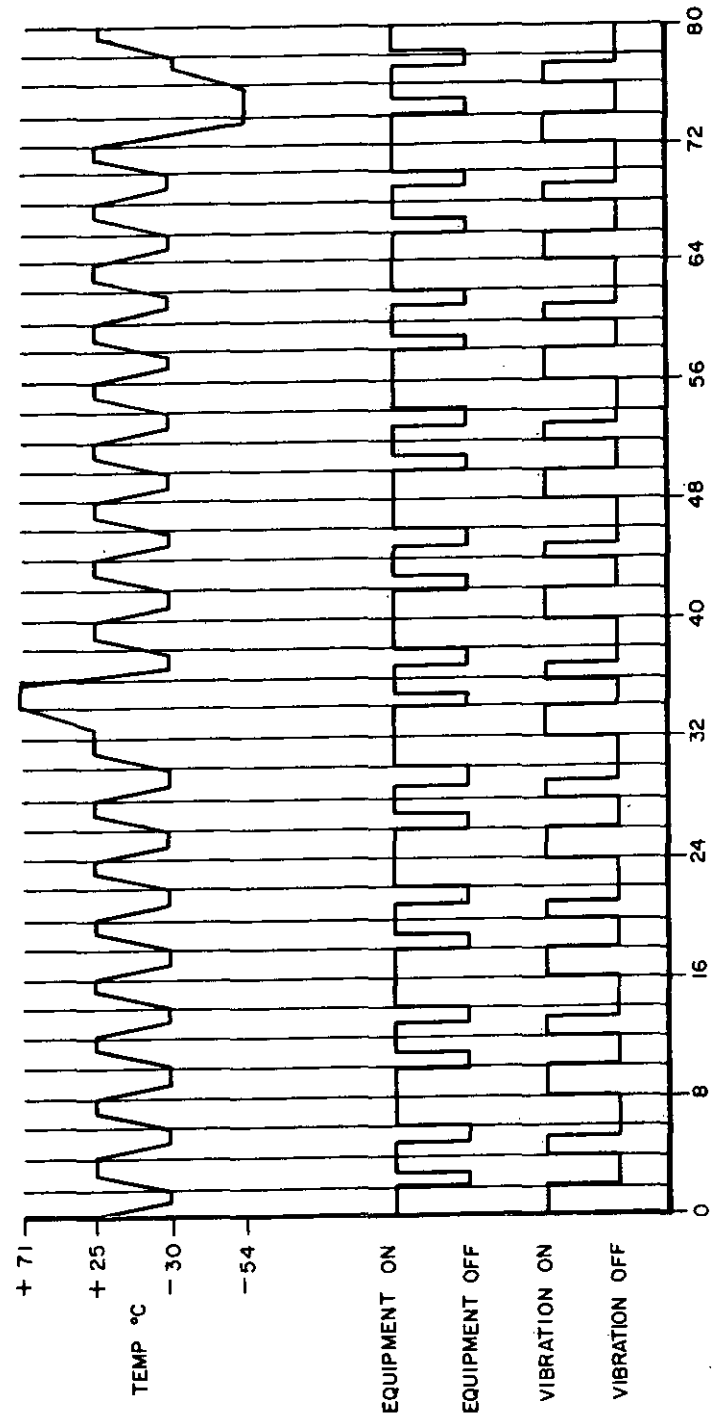
4.6.24 Shock. The shock test shall be conducted in accordance with MIL-STD-810, method 516, procedures I and III. Following the test specified in procedure I, the gyroscope assembly shall meet the individual tests except 4.6.2, 4.6.8, and 4.6.9.

4.6.25 Vibration failure. The gyroscope assembly shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514, equipment category (b), curve D, except that at 1g, the frequency shall be extended to 0.40 inch double amplitude and shall be maintained at this amplitude until the frequency is reduced to 5 Hz. At the completion of this test, the gyroscope assembly shall meet the individual tests except 4.6.2, 4.6.8, and 4.6.9.

4.6.26 Acoustical noise. The gyroscope assembly shall be subjected to an acoustical noise test in accordance with MIL-STD-810, method 515, procedure I, category A. At the conclusion of this test, the gyroscope assembly shall pass the individual tests except 4.6.2, 4.6.8, and 4.6.9.

4.6.27 Reliability. The gyroscope assembly shall first pass the sampling plan A tests and shall then be subjected to a reliability test as follows: The gyroscope assembly shall be properly connected with rated power applied and placed in a test chamber which shall meet the environmental cycling conditions shown on figure 3. During the ON cycle, the gyroscope assembly shall be exercised by a 3-axis, 30°, total-movement, 20-cpm reversing scorsby action and by the vibration shown on figure 3. After each 160 hours of cycling, the gyroscope assembly shall pass the individual tests except 4.6.2, 4.6.9, and 4.6.11. The

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a. Temperature change 2°C/minute average. Periods of +71° and -54°C, test may be changed to coincide with work shift
b. Equipment-off period = 1/2 hour within the period shown
c. Vibration-on period - 3 hrs/8 hrs approx as shown
d. Temp chamber shall be stabilized at the temp shown at each hour point
e. Environment cycle repeats at each 80 hours
f. Vibration level = 0.5g 20 to 30 Hz
g. Monitoring of outputs for catastrophic failure each ON cycle
h. Monitoring of performance conducted each 160 hours
i. Accumulated hours = all operating time.

FIGURE 3. Reliability Test Environment Cycles

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tolerances shall be in accordance with 4.6.27.2. Upon completion of this test, the gyroscope assembly shall pass the sampling plan A tests. All failures shall be considered relevant unless approved for exclusion by the procuring activity.

4.6.27.1 Longevity. After completion of the test specified in 4.6.27, testing of three of the gyroscope assemblies shall be continued for a total of 3,000 hours each. The longevity test shall be conducted during preproduction testing only.

4.6.27.2 Tolerances. Allowable tolerances for the individual tests conducted during reliability and longevity testing shall be in accordance with table II.

TABLE II. Tolerances

Increase in Tolerance	Time Per Unit (Hours)	Tests
0	0 to 500	4.6.3 through 4.6.10
0	500 to 1,000	4.6.3, 4.6.4
25 percent of specified	1,000 to 2,000	4.6.4, 4.6.8, 4.6.9
50 percent of specified	2,000 to 3,000	4.6.4, 4.6.8, 4.6.9
75 percent of specified	After 3,000	4.6.4, 4.6.8, 4.6.9
0.1 percent	500 to 1,000	4.6.5
0.15	1,000 to 3,000	4.6.5
25 percent of specified	500 to 1,000	4.6.6, 4.6.7, 4.6.10
50 percent of specified	1,000 to 3,000	4.6.6, 4.6.7, 4.6.10
75 percent of specified	After 3,000	4.6.6, 4.6.7, 4.6.10

* 4.6.27.2.1 If the gyroscope assemblies pass the reliability test in less than 3,000 hours of operation, the following tolerances shall apply for the sampling plan A tests:

a. Individual tests: In accordance with table II

b. 4.6.12 and 4.6.13: Tolerances for 4.6.4 in accordance with table II

c. 4.6.10: No tolerance increase

d. 4.6.11: No tolerance increase

e. 4.6.14: Tolerances for 4.6.4 during voltage extremes shall be +12 and -20 percent

f. 4.6.15: No tolerance increase

g. 4.6.16: Output signal for roll, pitch, and yaw no greater than 30 mv, 40 mv, and 30 mv respectively

h. 4.6.17: No increase in tolerance.

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* 4.6.27.2.2 After 3,000 hours of gyroscope assembly operation, the sampling plan A test tolerances shall be as follows:

a. Individual tests: In accordance with table II

b. 4.6.12 and 4.6.13:

(1) Tolerance for 4.6.4 shall be increased by 100 percent of specified tolerance

(2) No tolerance increase

c. 4.6.11: No tolerance increase

d. 4.6.14: Tolerances for 4.6.4 during voltage extremes shall be +12 and -20 percent of normal output at the respective rate for each axis

e. 4.6.15: No increase in tolerance

f. 4.6.16: Output signal for roll, pitch, and yaw shall be no greater than 35 mv, 45 mv, and 35 mv respectively

g. 4.6.17: No increase in tolerance.

4.6.27.3 Test records. Throughout the reliability test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-STD-781: Operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the records shall be submitted to the procuring activity within 30 days after completion of each test phase.

4.6.27.4 Repair of failed units. In the event of a failure during testing, the failed unit shall be repaired and the test continued.

4.6.28 Maintainability demonstration. The gyroscope assembly shall be subjected to a maintainability demonstration test in accordance with MIL-STD-471, appendix B, test method 1, test plan A_1 plus B_1. The assembly shall have a M_{ct} of 100 minutes and a M_{maxct} of not more than 300 minutes less the time required to unseal and hermetically seal the case after repair.

* 4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected to determine compliance with section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing

5.1.1 Levels A, B, and C. Each gyroscope assembly shall be preserved in accordance with MIL-P-116, method IC, and packaged and packed in accordance with MIL-STD-794 utilizing a reusable container capable of providing adequate protection to the assembly against the induced and natural forces encountered in transportation and storage.

5.1.1.1 Container design. The container design shall be of the minimum weight and size commensurate with performance requirements. The container shall generally conform to the requirements of MIL-C-4150 for carrying cases except that the method of sealing and container accessory items shall be only those essential for water-proofness (in lieu of water-vaporproofness).

5.1.2 Support cushion. A shock-vibration damping system of molded polyurethane foam conforming to MIL-P-26514 shall be provided in accordance with appendix I of MIL-C-4150 to insure that transmission of these forces are less than the structural values established for the gyroscope assembly.

5.2 Tests. Rough handling tests and such other tests as are prescribed by MIL-STD-794 and MIL-P-116 shall be performed to substantiate the capability of the container and its cushioning to perform its intended mission. Test reports and drawings shall be submitted to the procuring activity for review and approval prior to container production.

5.3 Marking. In addition to the marking for shipment requirements of MIL-STD-129, the following marking shall be placed in a conspicuous location on two opposite sides of each container:

FRAGILE
DELICATE INSTRUMENT
HANDLE WITH EXTREME CARE

5.3.1 In addition, each container shall be conspicuously marked on two opposite sides as follows: REUSABLE CONTAINER, USAF PROPERTY, OPEN WITH CARE.

6. NOTES

6.1 Intended use. The SBU-13/A gyroscope assembly covered by this specification is intended for use in providing pitch, roll, and yaw rates to an MXU-553/A signal data recorder.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. When sampling plan C test will not be conducted

d. Levels of packaging and packing required.

6.3 Test information. Sixty days prior to starting the testing, the contractor will submit detailed test information to the procuring activity for approval. This information should include but not necessarily be limited to:

a. Test procedures to be used

b. Equipment used in monitoring performance

c. Periodicity of calibration checks to be made on monitoring equipment

d. Description of environmental test facilities to be used, including recording facilities, methods of controlling heating and cooling facilities, and vibration equipment

e. Frequency of monitoring.

6.3.1 Failure report. The contractor will submit a report of all failures that occur during testing. The report will be submitted within two working days.

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

6.4.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic, or both, materials. A hermetic seal does not include seals accomplished by gaskets.

6.4.2 Failure. Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

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6.5 Marginal indicia. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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